Math 5312 / Math 7363 Midterm

Spring 2026

Student Name: _____

Submission Deadline: March 31, 2026, during class, **or** by email no later than 11:59 PM on March 31, 2026.

Instructions

1. You must submit your solutions written by hand.
2. Typeset LaTeX submissions are **not** allowed.
3. You may write on paper or on a tablet/iPad using a stylus, but only handwritten work is permitted.
4. You are encouraged to discuss the problems with others, but each submitted solution must be written independently.
5. Show all essential steps of your reasoning.
6. Clearly state any theorems you use.
7. Organize your work carefully and label each problem clearly.

Problem 1: Differential Forms and Vector Calculus on $\mathbb{R}^3$

In this problem you will study differential forms on $\mathbb{R}^3$ and their relationship with classical vector calculus operators.

Coordinates
We work on

$$\mathbb{R}^3$$

with coordinates

$$(x, y, z).$$

Differential Forms

- A **0-form** is a smooth function

$$f(x, y, z).$$

  The vector space of all 0-forms is denoted

$$\Omega^0(\mathbb{R}^3).$$

- A **1-form** is an expression of the form

$$\omega = a\,dx + b\,dy + c\,dz$$

  where a, b, c are smooth functions.

  The vector space of 1-forms is denoted

$$\Omega^1(\mathbb{R}^3).$$

- A **2-form** is an expression of the form

$$\eta = p\,dy \wedge dz + q\,dz \wedge dx + r\,dx \wedge dy.$$

  The vector space of 2-forms is denoted

$$\Omega^2(\mathbb{R}^3).$$

- A **3-form** has the form

$$\alpha = g\,dx \wedge dy \wedge dz.$$

The vector space of 3-forms is denoted

$$\Omega^3(\mathbb{R}^3).$$

Wedge Product

The wedge product $\wedge$ is an antisymmetric product defined by

$$dx \wedge dy = -dy \wedge dx, \qquad dx \wedge dx = 0.$$

For example

$$(a\,dx + b\,dy) \wedge (c\,dx + d\,dy) = \begin{vmatrix} a & b \\ c & d \end{vmatrix} dx \wedge dy.$$

Thus the wedge product behaves like a determinant.

Exterior Derivative

The exterior derivative

$$d : \Omega^k(\mathbb{R}^3) \to \Omega^{k+1}(\mathbb{R}^3)$$

is defined as follows.

For a function f,

$$df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy + \frac{\partial f}{\partial z}dz.$$

For a 1-form

$$\omega = a\,dx + b\,dy + c\,dz$$

$$d\omega = (\partial_x b - \partial_y a)\,dx \wedge dy + (\partial_y c - \partial_z b)\,dy \wedge dz + (\partial_z a - \partial_x c)\,dz \wedge dx.$$

For a 2-form

$$\eta = p\,dy \wedge dz + q\,dz \wedge dx + r\,dx \wedge dy$$

$$d\eta = (\partial_x p + \partial_y q + \partial_z r)\,dx \wedge dy \wedge dz.$$

Thus we obtain a sequence of vector spaces

$$\Omega^0(\mathbb{R}^3) \xrightarrow{d} \Omega^1(\mathbb{R}^3) \xrightarrow{d} \Omega^2(\mathbb{R}^3) \xrightarrow{d} \Omega^3(\mathbb{R}^3).$$

Vector Calculus Operators
For a scalar function f,

$$\nabla f = (\partial_x f, \partial_y f, \partial_z f).$$

For a vector field

$$F = (F_1, F_2, F_3),$$

$$\nabla \times F = (\partial_y F_3 - \partial_z F_2, \partial_z F_1 - \partial_x F_3, \partial_x F_2 - \partial_y F_1).$$

$$\nabla \cdot F = \partial_x F_1 + \partial_y F_2 + \partial_z F_3.$$

(a) Gradient and 1-Forms

Let

$$f(x, y, z) = x^2 y + y z^2.$$

1. Compute the exterior derivative df.

2. Compute the gradient vector field

$$\nabla f.$$

3. Show explicitly that the coefficients of df correspond to the components of ∇f.

4. Explain how this gives a correspondence between the vector spaces

$$\Omega^0(\mathbb{R}^3) \quad \text{and} \quad \Omega^1(\mathbb{R}^3).$$

(b) Vector Fields and 1-Forms

Let

$$F = (F_1, F_2, F_3)$$

be a vector field.

1. Construct the corresponding 1-form

$$\omega = F_1 dx + F_2 dy + F_3 dz.$$

2. Show that this construction gives a one-to-one correspondence between vector fields on $\mathbb{R}^3$ and 1-forms.

(c) Curl and 2-Forms

Let

$$F = (F_1, F_2, F_3)$$

and

$$\omega = F_1 dx + F_2 dy + F_3 dz.$$

1. Compute the exterior derivative $d\omega$.

2. Compute the curl

$$\nabla \times F.$$

3. Show that the coefficients of the 2-form $d\omega$ correspond to the components of $\nabla \times F$.

4. Explain how this gives a correspondence between

$$\Omega^1(\mathbb{R}^3) \quad \text{and} \quad \Omega^2(\mathbb{R}^3).$$

(d) Divergence and 3-Forms

Let

$$F = (F_1, F_2, F_3)$$

and define the 2-form

$$\eta = F_1\, dy \wedge dz + F_2\, dz \wedge dx + F_3\, dx \wedge dy.$$

1. Compute $d\eta$.

2. Compute

$$\nabla \cdot F.$$

3. Show that the coefficient of the 3-form $d\eta$ equals $\nabla \cdot F$.

4. Explain how this produces a correspondence between

$$\Omega^2(\mathbb{R}^3) \quad \text{and} \quad \Omega^3(\mathbb{R}^3).$$

(e) Summary

Using the previous parts, explain the correspondence

$$\Omega^0 \xrightarrow{d} \Omega^1 \xrightarrow{d} \Omega^2 \xrightarrow{d} \Omega^3$$

with the vector calculus operators

$$\text{grad}, \quad \text{curl}, \quad \text{div}.$$

Problem 2: Computing the Winding 1-Form on the Unit Circle

Consider the punctured plane

$$\mathbb{R}^2 \setminus \{0\}$$

with coordinates

$$(x, y).$$

Define the winding 1-form

$$\omega = \frac{-y\,dx + x\,dy}{x^2 + y^2}.$$

Let

$$S^1 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$$

be the unit circle.
The goal of this problem is to compute

$$\int_{S^1} \omega$$

using three different approaches.

(a) Angular Parameterization

Use the parameterization

$$x = \cos\theta, \qquad y = \sin\theta, \qquad 0 \le \theta \le 2\pi.$$

Compute the integral

$$\int_{S^1} \omega.$$

(b) Computation Using the Relation $x^2 + y^2 = 1$

Use the identity

$$x^2 + y^2 = 1$$

on the unit circle and differentiate it to obtain a relation between dx and dy. Use this relation to simplify the winding form

$$\omega = \frac{-y\,dx + x\,dy}{x^2 + y^2}$$

on S^1 and compute the integral

$$\int_{S^1} \omega.$$

(c) Complex Variable Method

Introduce the complex coordinate

$$z = x + iy.$$

Rewrite the winding form using

$$\frac{dz}{z}.$$

Then use the parameterization

$$z = e^{i\theta}$$

for the unit circle and compute

$$\int_{S^1} \omega.$$

Problem 3: Cauchy–Green Formula via the Generalized Stokes Theorem

In this problem you will derive the Cauchy–Green formula using the generalized Stokes theorem. The setup follows the argument developed in the lecture notes.

Let

$$\Omega \subset \mathbb{C}$$

be a bounded domain with smooth boundary $\partial\Omega$ and suppose

$$f \in C^1(\overline{\Omega}).$$

Fix

$$z \in \Omega$$

and choose $\varepsilon > 0$ small enough so that

$$B_\varepsilon(z) \subset \Omega.$$

Define

$$\Omega_\varepsilon = \Omega \setminus B_\varepsilon(z).$$

Define the 1-form

$$\alpha(\omega) = \frac{f(\omega)}{\omega - z}\, d\omega.$$

(a) Boundary of the punctured domain

Show that the boundary of Ω_ε is

$$\partial\Omega_\varepsilon = \partial\Omega \cup (-\partial B_\varepsilon(z)).$$

Write the integral

$$\int_{\partial\Omega_\varepsilon} \alpha$$

as the difference of two boundary integrals.

(b) Exterior derivative of the 1-form

Let

$$g(\omega) = \frac{f(\omega)}{\omega - z}.$$

Compute

$$d\alpha = d(g\, d\omega).$$

Use the identity

$$d(g\, d\omega) = dg \wedge d\omega.$$

Write

$$dg = \frac{\partial g}{\partial \omega} d\omega + \frac{\partial g}{\partial \overline{\omega}} d\overline{\omega}.$$

Show that

$$d\alpha = \frac{\partial g}{\partial \overline{\omega}}\, d\overline{\omega} \wedge d\omega.$$

(c) Computing the derivative

Show that

$$\frac{\partial g}{\partial \overline{\omega}} = \frac{1}{\omega - z} \frac{\partial f}{\partial \overline{\omega}}.$$

Deduce that

$$d\alpha = \frac{\partial f / \partial \overline{\omega}}{\omega - z}\, d\overline{\omega} \wedge d\omega.$$

(d) Apply the generalized Stokes theorem

Apply the generalized Stokes theorem

$$\int_{\partial\Omega_\varepsilon} \alpha = \iint_{\Omega_\varepsilon} d\alpha.$$

Write the result as

$$\int_{\partial\Omega} \frac{f(\omega)}{\omega - z} d\omega - \int_{\partial B_\varepsilon(z)} \frac{f(\omega)}{\omega - z} d\omega = \iint_{\Omega_\varepsilon} \frac{\partial f / \partial \overline{\omega}}{\omega - z} \, d\overline{\omega} \wedge d\omega.$$

(e) Limit of the small circle integral

Show that

$$\lim_{\varepsilon \to 0} \int_{\partial B_\varepsilon(z)} \frac{f(\omega)}{\omega - z} d\omega = 2\pi i \, f(z).$$

(f) Cauchy–Green formula

Taking the limit $\varepsilon \to 0$, deduce the formula

$$f(z) = \frac{1}{2\pi i} \int_{\partial\Omega} \frac{f(\omega)}{\omega - z} d\omega + \frac{1}{2\pi i} \iint_{\Omega} \frac{\partial f / \partial \overline{\omega}}{\omega - z} \, d\overline{\omega} \wedge d\omega.$$

Explain why this is the Cauchy–Green formula.

Problem 4: Classical Integral Theorems and Differential Forms

In this problem you will apply several classical integral theorems from vector calculus and compare them with their formulations using differential forms.

Throughout we work with smooth functions and smooth domains.

(a) Fundamental Theorem of Line Integrals

Classical Statement

Let $f : \mathbb{R}^n \to \mathbb{R}$ be a smooth function and let $\gamma : [a, b] \to \mathbb{R}^n$ be a smooth curve.
Then

$$\int_\gamma \nabla f \cdot d\mathbf{r} = f(\gamma(b)) - f(\gamma(a)).$$

Differential Form Version

If

$$df = \sum_i \frac{\partial f}{\partial x_i} dx_i,$$

then

$$\int_\gamma df = f(\gamma(b)) - f(\gamma(a)).$$

Problem

Let

$$f(x, y) = x^2 y.$$

Let $\gamma(t) = (t, t^2)$ for $0 \le t \le 1$.

1. Compute the gradient ∇f.

2. Compute the line integral

$$\int_\gamma \nabla f \cdot d\mathbf{r}.$$

3. Compute $f(\gamma(1)) - f(\gamma(0))$ and verify the theorem.

(b) Green's Theorem

Classical Statement

Let $D \subset \mathbb{R}^2$ be a region with positively oriented boundary ∂D.

Then

$$\oint_{\partial D} (P\,dx + Q\,dy) = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx\,dy.$$

Differential Form Version

Let

$$\omega = P\,dx + Q\,dy.$$

Then

$$\int_{\partial D} \omega = \int_D d\omega.$$

Problem

Let

$$P(x, y) = -y, \qquad Q(x, y) = x.$$

Let D be the unit disk.

1. Compute

$$\oint_{\partial D} (P\,dx + Q\,dy).$$

2. Compute

$$\iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx\, dy.$$

3. Verify Green's theorem.

(c) Classical Stokes Theorem

Classical Statement

Let S be an oriented surface with boundary ∂S and let F be a vector field.
Then

$$\oint_{\partial S} F \cdot d\mathbf{r} = \iint_S (\nabla \times F) \cdot n \, dS.$$

Differential Form Version

Let

$$\omega = F_1 dx + F_2 dy + F_3 dz.$$

Then

$$\int_{\partial S} \omega = \int_S d\omega.$$

Problem

Let

$$F = (-y, x, 0).$$

Let S be the unit disk in the plane $z = 0$.

1. Compute $\nabla \times F$.

2. Compute the surface integral

$$\iint_S (\nabla \times F) \cdot n \, dS.$$

3. Compute the boundary integral

$$\oint_{\partial S} F \cdot d\mathbf{r}.$$

4. Verify Stokes' theorem.

(d) Divergence Theorem

Classical Statement

Let $V \subset \mathbb{R}^3$ be a region with boundary ∂V.

Then

$$\iiint_V (\nabla \cdot F) dV = \iint_{\partial V} F \cdot n \, dS.$$

Differential Form Version

Let

$$\eta = F_1 \, dy \wedge dz + F_2 \, dz \wedge dx + F_3 \, dx \wedge dy.$$

Then

$$\int_{\partial V} \eta = \int_V d\eta.$$

Problem

Let

$$F = (x, y, z).$$

Let V be the unit ball.

1. Compute the divergence

$$\nabla \cdot F.$$

2. Compute the volume integral

$$\iiint_V (\nabla \cdot F) dV.$$

3. Compute the flux

$$\iint_{\partial V} F \cdot n \, dS.$$

4. Verify the divergence theorem.

Problem 5: From $\bar{z}$-Derivatives to the Cauchy Integral Theorem

In this problem you will show that holomorphicity implies the Cauchy–Riemann equations in the form

$$\frac{\partial f}{\partial \bar{z}} = 0,$$

and then use this fact to derive the Cauchy Integral Theorem.

Definitions

Let

$$z = x + iy, \qquad \bar{z} = x - iy.$$

If

$$f(z) = u(x, y) + iv(x, y),$$

where u, v are real-valued functions, define

$$\frac{\partial}{\partial z} = \frac{1}{2}\left(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\right), \qquad \frac{\partial}{\partial \bar{z}} = \frac{1}{2}\left(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\right).$$

Thus

$$\frac{\partial f}{\partial \bar{z}} = \frac{1}{2}\left(\frac{\partial f}{\partial x} + i\frac{\partial f}{\partial y}\right).$$

A function f is **holomorphic** if the complex derivative

$$f'(z) = \lim_{h \to 0} \frac{f(z + h) - f(z)}{h}$$

exists.

Green's Theorem

Let $D \subset \mathbb{R}^2$ be a bounded region with positively oriented smooth boundary ∂D. If $P, Q \in C^1(\overline{D})$, then

$$\oint_{\partial D} (P\,dx + Q\,dy) = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy.$$

Cauchy Integral Theorem (CIT)

If f is holomorphic on a domain containing a bounded region $D \subset \mathbb{C}$ and its boundary ∂D, then

$$\oint_{\partial D} f(z)\,dz = 0.$$

(a) Compute $\partial f/\partial \bar{z}$ in terms of u and v

Let

$$f(z) = u(x,y) + iv(x,y).$$

Compute

$$\frac{\partial f}{\partial x}, \qquad \frac{\partial f}{\partial y},$$

and then show that

$$\frac{\partial f}{\partial \bar{z}} = \frac{1}{2}\Big[\big(u_x - v_y\big) + i\big(v_x + u_y\big)\Big].$$

(b) Show that $\partial f/\partial \bar{z} = 0$ is equivalent to the Cauchy–Riemann equations

Show that

$$\frac{\partial f}{\partial \bar{z}} = 0$$

if and only if

$$u_x = v_y, \qquad u_y = -v_x.$$

Explain why these are exactly the Cauchy–Riemann equations.

(c) Express $f(z)\,dz$ in real coordinates

Write

$$dz = dx + i\,dy.$$

Show that

$$f(z)\,dz = (u+iv)(dx+i\,dy) = (u\,dx - v\,dy) + i(v\,dx + u\,dy).$$

Deduce that

$$\oint_{\partial D} f(z)\,dz = \oint_{\partial D} (u\,dx - v\,dy) + i \oint_{\partial D} (v\,dx + u\,dy).$$

(d) Apply Green's theorem

Apply Green's theorem separately to the two real integrals in part (c).

Show that

$$\oint_{\partial D} (u\,dx - v\,dy) = \iint_D \left(\frac{\partial(-v)}{\partial x} - \frac{\partial u}{\partial y} \right) dx\,dy$$

and

$$\oint_{\partial D} (v\,dx + u\,dy) = \iint_D \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \right) dx\,dy.$$

Problem 6: Gaussian Curvature, Area Forms, and Euler Characteristic

In this problem you will compute the Gaussian curvature and area form for two surfaces in $\mathbb{R}^3$:

$$S^2 \quad \text{and} \quad T^2$$

and verify directly that

$$\int_M K \, dA = 2\pi\chi(M).$$

You will also verify that the surface area element obtained from a parametrization agrees with the area obtained by a change of variables.

Surface geometry formulas
Let

$$X(u,v) : U \subset \mathbb{R}^2 \to \mathbb{R}^3$$

be a parametrized surface.
Define

$$E = \langle X_u, X_u \rangle, \qquad F = \langle X_u, X_v \rangle, \qquad G = \langle X_v, X_v \rangle.$$

The area form is

$$dA = \sqrt{EG - F^2}\, du\, dv.$$

If N is a unit normal field, define

$$e = \langle X_{uu}, N \rangle, \qquad f = \langle X_{uv}, N \rangle, \qquad g = \langle X_{vv}, N \rangle.$$

The Gaussian curvature is

$$K = \frac{eg - f^2}{EG - F^2}.$$

The Gauss–Bonnet formula for a closed surface M is

$$\int_M K \, dA = 2\pi\chi(M).$$

Problem (a): Geometry of the sphere in spherical coordinates

In this problem you will use spherical coordinates to compute the area element, the Gaussian curvature, and the Euler characteristic of the unit sphere

$$S^2 \subset \mathbb{R}^3.$$

Consider the standard spherical parametrization

$$X(\phi, \theta) = (\sin\phi\cos\theta,\ \sin\phi\sin\theta,\ \cos\phi),$$

where

$$0 < \phi < \pi, \qquad 0 < \theta < 2\pi.$$

Recall the following formulas for a parametrized surface:

$$E = \langle X_\phi, X_\phi \rangle, \qquad F = \langle X_\phi, X_\theta \rangle, \qquad G = \langle X_\theta, X_\theta \rangle,$$

$$dA = \sqrt{EG - F^2}\, d\phi\, d\theta,$$

and, for a unit normal field N,

$$e = \langle X_{\phi\phi}, N \rangle, \qquad f = \langle X_{\phi\theta}, N \rangle, \qquad g = \langle X_{\theta\theta}, N \rangle.$$

The Gaussian curvature is

$$K = \frac{eg - f^2}{EG - F^2}.$$

1. Compute the tangent vectors

$$X_\phi, \qquad X_\theta.$$

2. Compute the coefficients of the first fundamental form:

$$E = \langle X_\phi, X_\phi \rangle, \qquad F = \langle X_\phi, X_\theta \rangle, \qquad G = \langle X_\theta, X_\theta \rangle.$$

3. Deduce the spherical area element

$$dA = \sqrt{EG - F^2}\, d\phi\, d\theta$$

and simplify it to the standard form.

4. Compute the total area of the unit sphere by integrating the area element over the full range of spherical coordinates.

5. Compute the outward unit normal vector field N. Show that for the unit sphere one may take

$$N = X.$$

6. Compute the second derivatives

$$X_{\phi\phi}, \qquad X_{\phi\theta}, \qquad X_{\theta\theta}.$$

7. Compute the coefficients of the second fundamental form:

$$e = \langle X_{\phi\phi}, N \rangle, \qquad f = \langle X_{\phi\theta}, N \rangle, \qquad g = \langle X_{\theta\theta}, N \rangle.$$

8. Use the formula

$$K = \frac{eg - f^2}{EG - F^2}$$

to compute the Gaussian curvature K of the unit sphere.

9. Compute the total curvature

$$\int_{S^2} K \, dA = \int_0^{2\pi} \int_0^{\pi} K(\phi, \theta) \sqrt{EG - F^2} \, d\phi \, d\theta.$$

10. Since the Gauss–Bonnet formula for a closed surface is

$$\int_M K \, dA = 2\pi\chi(M),$$

use your answer from part (9) to determine the Euler characteristic

$$\chi(S^2).$$

11. Explain geometrically why the positivity and constancy of the Gaussian curvature of the sphere are compatible with the fact that

$$\chi(S^2) = 2.$$

Problem (b): Area form and change of variables

Let

$$x = \sin\phi\cos\theta, \qquad y = \sin\phi\sin\theta.$$

1. Compute the Jacobian

$$\frac{\partial(x,y)}{\partial(\phi,\theta)}.$$

2. Show that the area form

$$dA = \sin\phi\, d\phi\, d\theta$$

agrees with the area obtained from the change of variables formula

$$dx\, dy = \left|\det \frac{\partial(x,y)}{\partial(\phi,\theta)}\right| d\phi\, d\theta.$$

3. Explain why this confirms that the geometric area of the sphere does not depend on the chosen coordinates.

Problem (c): Geometry of the torus

Let $R > r > 0$. Consider the standard torus

$$T^2$$

parametrized by

$$X(u, v) = ((R + r\cos u)\cos v, (R + r\cos u)\sin v, r\sin u),$$

$$0 < u < 2\pi, \quad 0 < v < 2\pi.$$

1. Compute E, F, G and deduce the area form dA.

2. Compute the Gaussian curvature and show that

$$K(u, v) = \frac{\cos u}{r(R + r\cos u)}.$$

3. Compute

$$\int_{T^2} K \, dA.$$

4. Verify that

$$\int_{T^2} K \, dA = 2\pi\chi(T^2).$$

Problem 7: Euler Characteristic of $\mathbb{CP}^1$ and a Complex Torus

In this problem you will compute the Euler characteristic using complex coordinates and differential forms.

The goal is to verify directly that

$$\chi(\mathbb{CP}^1) = 2, \qquad \chi(T^2) = 0.$$

Recall that for a closed oriented surface M

$$\chi(M) = \frac{1}{2\pi} \int_M K \, dA.$$

You will compute the curvature and area form using complex coordinates.

Useful formulas

Let $z = x + iy$. The complex derivatives are

$$\partial = \frac{1}{2}\left(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\right), \qquad \bar{\partial} = \frac{1}{2}\left(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\right).$$

If a Hermitian metric is written in local coordinates as

$$ds^2 = \lambda(z, \bar{z})\, |dz|^2,$$

then the Gaussian curvature is

$$K = -\frac{1}{\lambda}\partial\bar{\partial} \log \lambda.$$

The associated area form is

$$dA = \lambda \, dx \, dy = \frac{i}{2}\lambda \, dz \wedge d\bar{z}.$$

(a) Complex coordinates on $\mathbb{CP}^1$

The Riemann sphere $\mathbb{CP}^1$ can be covered by the coordinate

$$z \in \mathbb{C}$$

obtained from stereographic projection.
The standard Fubini–Study metric is

$$ds^2 = \frac{4}{(1+|z|^2)^2}|dz|^2.$$

Compute the corresponding area form.

(b) Gaussian curvature of $\mathbb{CP}^1$

Let

$$\lambda(z, \bar{z}) = \frac{4}{(1 + |z|^2)^2}.$$

Compute

$$\partial\bar{\partial} \log \lambda.$$

Use the curvature formula

$$K = -\frac{1}{\lambda}\partial\bar{\partial} \log \lambda$$

to show that

$$K = 1.$$

(c) Euler characteristic of $\mathbb{CP}^1$

Using the area form from part (a) and the curvature from part (b), compute

$$\int_{\mathbb{CP}^1} K \, dA.$$

Deduce that

$$\chi(\mathbb{CP}^1) = 2.$$

(d) Complex torus

Let

$$T^2 = \mathbb{C}/\Lambda$$

be a complex torus, where $\Lambda \subset \mathbb{C}$ is a lattice.
The torus inherits the flat metric

$$ds^2 = |dz|^2.$$

Compute the corresponding area form.

(e) Gaussian curvature of the complex torus

For the metric

$$ds^2 = |dz|^2$$

we have

$$\lambda = 1.$$

Compute

$$\partial\bar{\partial}\log\lambda$$

and deduce that

$$K = 0.$$

Problem 8: Partition of Unity on S^2 and Integration

In this problem you will use a partition of unity to compute an integral on the sphere S^2.

Definition (Partition of Unity). Let M be a smooth manifold and let $\{U_\alpha\}_{\alpha\in A}$ be an open cover of M. A collection of smooth functions

$$\{\varphi_\alpha\}_{\alpha\in A}$$

is called a **partition of unity subordinate to** $\{U_\alpha\}_{\alpha\in A}$ if the following conditions hold:

1. For each α,
$$0 \le \varphi_\alpha \le 1.$$

2. The support of φ_α is contained in U_α:
$$\operatorname{supp}(\varphi_\alpha) \subset U_\alpha.$$

3. At each point of M,
$$\sum_\alpha \varphi_\alpha = 1.$$
Only finitely many terms are nonzero near each point.

Integration with a partition of unity. If ω is a top-degree form on M, then

$$\int_M \omega = \sum_\alpha \int_M \varphi_\alpha \omega = \sum_\alpha \int_{U_\alpha} \varphi_\alpha \omega.$$

Let

$$S^2 = \{(x,y,z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 1\}.$$

Consider the open cover

$$U_N = S^2 \setminus \{(0,0,1)\}, \qquad U_S = S^2 \setminus \{(0,0,-1)\}.$$

Define smooth functions on S^2 by

$$\varphi_N(x,y,z) = \frac{1-z}{2}, \qquad \varphi_S(x,y,z) = \frac{1+z}{2}.$$

You may use the stereographic coordinates

$$\pi_N : U_N \to \mathbb{R}^2, \qquad \pi_S : U_S \to \mathbb{R}^2$$

from the north and south poles. In either stereographic chart, the standard area form on S^2 is

$$dA = \frac{4}{(1+|w|^2)^2}\,du\,dv \qquad (w = u + iv).$$

Also, in the north stereographic coordinate w,

$$z = \frac{|w|^2 - 1}{|w|^2 + 1},$$

and in the south stereographic coordinate w,

$$z = \frac{1 - |w|^2}{1 + |w|^2}.$$

(a) Verifying the partition of unity

Show that

$$0 \le \varphi_N, \varphi_S \le 1, \qquad \varphi_N + \varphi_S = 1 \quad \text{on } S^2.$$

Explain why $\{\varphi_N, \varphi_S\}$ is a partition of unity subordinate to $\{U_N, U_S\}$.

(b) Local expressions in stereographic coordinates

Using the north stereographic chart, express φ_N in terms of $w = u + iv$.

Show that

$$\varphi_N = \frac{1}{1 + |w|^2}.$$

Using the south stereographic chart, express φ_S in terms of $w = u + iv$ and show that

$$\varphi_S = \frac{1}{1 + |w|^2}.$$

(c) Integrating one term of the partition

Compute

$$\int_{U_N} \varphi_N \, dA$$

using north stereographic coordinates. Reduce the integral to an integral over $\mathbb{R}^2$, then use polar coordinates

$$u = r\cos\theta, \qquad v = r\sin\theta$$

to evaluate it explicitly.

(d) Integrating the second term

Compute

$$\int_{U_S} \varphi_S \, dA$$

using south stereographic coordinates.

(e) Recovering the total area of the sphere

Using the partition of unity identity

$$\int_{S^2} dA = \int_{U_N} \varphi_N \, dA + \int_{U_S} \varphi_S \, dA,$$

compute

$$\int_{S^2} dA.$$

Verify that the answer is

$$\text{Area}(S^2) = 4\pi.$$

Problem 9: Mayer–Vietoris Sequence and de Rham Cohomology

In this problem you will derive the Mayer–Vietoris sequence for de Rham cohomology and verify how a short exact sequence of differential forms produces a long exact sequence in cohomology.

Both the problems and detailed solutions are included.

Definition (de Rham complex).
For a smooth manifold M, the spaces of differential forms

$$\Omega^0(M), \Omega^1(M), \Omega^2(M), \dots$$

together with the exterior derivative

$$d : \Omega^k(M) \to \Omega^{k+1}(M)$$

form a complex

$$0 \to \Omega^0(M) \xrightarrow{d} \Omega^1(M) \xrightarrow{d} \Omega^2(M) \xrightarrow{d} \cdots$$

satisfying

$$d^2 = 0.$$

Definition (Closed and exact forms).
A k-form ω is

$$\text{closed} \quad \text{if} \quad d\omega = 0$$

and

$$\text{exact} \quad \text{if} \quad \omega = d\eta$$

for some $(k-1)$-form η.

Definition (de Rham cohomology).
The k-th de Rham cohomology group of M is

$$H^k_{dR}(M) = \frac{\ker(d : \Omega^k(M) \to \Omega^{k+1}(M))}{\mathrm{Im}(d : \Omega^{k-1}(M) \to \Omega^k(M))}.$$

Thus

$$H^k_{dR}(M) = \frac{\text{closed } k\text{-forms}}{\text{exact } k\text{-forms}}.$$

(a) Short exact sequence of differential forms

Let $M = U \cup V$ be an open cover.

Define the restriction maps

$$i : \Omega^k(M) \to \Omega^k(U) \oplus \Omega^k(V), \qquad i(\omega) = (\omega|_U, \omega|_V)$$

and

$$\rho : \Omega^k(U) \oplus \Omega^k(V) \to \Omega^k(U \cap V)$$

by

$$\rho(\alpha, \beta) = \alpha|_{U\cap V} - \beta|_{U\cap V}.$$

Show that

$$0 \to \Omega^k(M) \to \Omega^k(U) \oplus \Omega^k(V) \to \Omega^k(U \cap V) \to 0$$

is exact.

Solution.

- Injectivity:

  If

  $$i(\omega) = (0, 0)$$

  then

  $$\omega|_U = 0, \qquad \omega|_V = 0.$$

  Since $U \cup V = M$, we obtain

  $$\omega = 0.$$

  Thus i is injective.

- Kernel computation:

  Suppose

  $$\rho(\alpha, \beta) = 0.$$

  Then

$$\alpha|_{U\cap V} = \beta|_{U\cap V}.$$

Therefore the forms α and β agree on the overlap and glue to produce a global form

$$\omega \in \Omega^k(M)$$

with

$$\omega|_U = \alpha, \qquad \omega|_V = \beta.$$

Hence

$$(\alpha, \beta) = i(\omega).$$

Thus

$$\ker(\rho) = \mathrm{Im}(i).$$

- Surjectivity:

  Let $\gamma \in \Omega^k(U \cap V)$.

  Choose a partition of unity $\{\phi_U, \phi_V\}$ subordinate to the cover $\{U, V\}$.

  Define

  $$\alpha = \phi_V \, \gamma \quad \text{on } U$$

  and

  $$\beta = -\phi_U \, \gamma \quad \text{on } V.$$

  Then

  $$\rho(\alpha, \beta) = \gamma.$$

  Thus the map is surjective.

Therefore the sequence is a short exact sequence.

(b) Compatibility with the exterior derivative

Show that the exterior derivative commutes with restriction maps.

Solution.
For any form ω,

$$(d\omega)|_U = d(\omega|_U).$$

Similarly

$$(d\omega)|_V = d(\omega|_V).$$

Thus the maps in the short exact sequence commute with d.
Therefore we obtain a short exact sequence of complexes.

(c) Constructing the connecting homomorphism

Explain how the connecting homomorphism

$$\delta : H^k(U \cap V) \to H^{k+1}(M)$$

is constructed.

Solution.
Let

$$[\gamma] \in H^k(U \cap V)$$

so

$$d\gamma = 0.$$

Using surjectivity of the previous sequence, choose

$$(\alpha, \beta)$$

with

$$\rho(\alpha, \beta) = \gamma.$$

Compute

$$d\alpha, \quad d\beta.$$

These agree on $U \cap V$, so they glue to produce a global form

$$\omega \in \Omega^{k+1}(M).$$

Since $d^2 = 0$, the form ω is closed.
Define

$$\delta([\gamma]) = [\omega] \in H^{k+1}(M).$$

(d) The Mayer–Vietoris long exact sequence via the Snake Lemma

In this part we explain how the long exact sequence arises from the short exact sequence of complexes using the **Snake Lemma**. We first recall the algebraic version for modules and then apply it to the de Rham complex.

Algebraic setting (modules).
Let R be a ring and consider a commutative diagram of R-modules

$$\begin{array}{ccccccccc} 0 & \to & A & \xrightarrow{i} & B & \xrightarrow{p} & C & \to & 0 \\ & & \downarrow f & & \downarrow g & & \downarrow h & & \\ 0 & \to & A' & \xrightarrow{i'} & B' & \xrightarrow{p'} & C' & \to & 0 \end{array}$$

in which both rows are short exact sequences.

Snake Lemma.
From this diagram there exists a connecting homomorphism

$$\delta : \ker(h) \to \operatorname{coker}(f)$$

and an exact sequence

$$\ker(f) \to \ker(g) \to \ker(h) \xrightarrow{\delta} \operatorname{coker}(f) \to \operatorname{coker}(g) \to \operatorname{coker}(h).$$

This result is known as the **Snake Lemma**. It is one of the basic tools used to produce long exact sequences in homological algebra.

(e) Applying the Snake Lemma to the de Rham complex

Recall from the previous part that we have a short exact sequence of complexes

$$0 \to \Omega^\bullet(M) \to \Omega^\bullet(U) \oplus \Omega^\bullet(V) \to \Omega^\bullet(U \cap V) \to 0.$$

At each degree k this gives the short exact sequence

$$0 \to \Omega^k(M) \to \Omega^k(U) \oplus \Omega^k(V) \to \Omega^k(U \cap V) \to 0.$$

Since the exterior derivative commutes with restriction maps, we obtain a commutative diagram

$$\begin{array}{ccccccccc} 0 & \to & \Omega^k(M) & \to & \Omega^k(U) \oplus \Omega^k(V) & \to & \Omega^k(U \cap V) & \to & 0 \\ & & \downarrow d & & \downarrow d & & \downarrow d & & \\ 0 & \to & \Omega^{k+1}(M) & \to & \Omega^{k+1}(U) \oplus \Omega^{k+1}(V) & \to & \Omega^{k+1}(U \cap V) & \to & 0 \end{array}$$

Applying the Snake Lemma to this diagram produces a connecting homomorphism

$$\delta : \ker(d : \Omega^k(U \cap V) \to \Omega^{k+1}(U \cap V)) \to \frac{\Omega^{k+1}(M)}{\operatorname{Im}(d)}.$$

But

$$\ker(d) = Z^k, \qquad \Omega^{k+1}/\operatorname{Im}(d) = H^{k+1},$$

so this map becomes

$$\delta : H^k(U \cap V) \to H^{k+1}(M).$$

(f) The Mayer–Vietoris long exact sequence

Combining the maps obtained above yields the sequence

$$\cdots \to H^k(M) \to H^k(U) \oplus H^k(V) \to H^k(U \cap V) \xrightarrow{\delta} H^{k+1}(M) \to H^{k+1}(U) \oplus H^{k+1}(V) \to \cdots$$

This is the **Mayer–Vietoris long exact sequence** in de Rham cohomology.

Interpretation.
The logical structure is

Short exact sequence of complexes

$\Downarrow$

Snake Lemma

$\Downarrow$

Long exact sequence in cohomology.

Thus the Mayer–Vietoris sequence is an example of a general phenomenon in homological algebra: a short exact sequence of chain complexes produces a long exact sequence in cohomology.